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22 May 1985
SUPERSEDING
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MILITARY SPECIFICATION

SHOPS, ELECTRONIC, SHELTER MOUNTED AN/ASM-146( ) AND AN/ASM-147( )

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the following:

1.1.1 Description. Shop, Electronic, Shelter Mounted AN/ASM-146( ), hereinafter referred to as an Electronic Shop, is a mobile assemblage housed in a modified S-280( )/G shelter equipped with work benches, storage cabinets and racks, lighting, air conditioning, and power to form an electronic repair shop for the maintenance of aircraft electronic equipment. It is capable of being transported by railroad, military truck and airlift.

1.1.2 Description. Shop, Electronic, Shelter Mounted AN/ASM-147( ), hereinafter referred to as an Electronic Shop, is a mobile assemblage housed in a modified S-280( )/G shelter equipped with storage cabinets and racks, lighting, exhaust blowers, and a dehumidifier to form a storage shelter to support the AN/ASM-146( ) repair shop assemblage. It is capable of being transported by railroad, military truck and airlift.

Beneficial comments (recommendations, additions, deletions) and pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: AMSEL-ED-SS, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4940

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2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards Of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form or a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cables Assemblies and Cord Assemblies (for use in Electronic and associated Equipment)
MIL-W-6858	Welding, Resistance; Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickle Alloys, Heat Resisting Alloys, and Titanium Alloys, Spot and Seam
MIL-T-7928	Terminals, Lug and Splice, Crimp Style, Copper
MIL-W-8611	Welding, Metal Arc and Gas, Steels and Corrosion and Heat Resistant Alloys, Process for
MIL-M-13231	Marking of Electronic items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-W-45205 .	Welding, Gas, Metal Arc and Gas, Tungsten-Arc, Aluminum Alloys
MIL-S-55507	Shelter, Electrical Equipment (With or Without Equipment) Packaging Of

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electrical Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipments
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-810	Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the estent specified herein.

DRAWINGS

ELECTRONICS COMMAND

SC-DL-577800C	Shop , Electronic, Shelter Mounted, AN/ASM-146( )
SC-DL-578200	Shop, Electronic, Shelter Mounted, AN/ASM-147( )

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish a first article unit of the AN/ASM-146( ) and of the AN/ASM-147( ) in accordance with 4.3.

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3.2 Construction. (See 3,20)

3.2.1 Shelters. Unless otherwise specified, the Shelter, Electrical Equipment S-280( )/G will be Government furnished.

3.2.2 Electronic shops installation.-

a. Shop, Electronic, Shelter Mounted AN/ASM-146( ) shall be constructed in accordance with drawings listed on SC-DL-57780C and the requirements of this specification, and any other requirements specified in the contract.

b. Shop, Electronic, Shelter Mounted AN/ASM-147( ) shall be constructed in accordance with drawings listed on SC-DL-57820C and the requirements of this specification, and any other requirements specified in the contract.

3.3 Seal, Water Vapor. All joints and edges of the framing for the blower exhausts, power entrance box, and air conditioner openings shall be sealed to provide a vapor barrier against the entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All the joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws and fasteners shall be coated by a one-part sealant, Silastic 738 as made by Dow Corning, or equal, using the manufacturer recommended primer. (See 3.20)

3.4 Cleaning. After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of MIL-F-14072. Cleaning processes shall have no deleterious effect, Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after assembly of the units. (See 3.23)

3.5 Finish. Each Electronic Shop shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 3.2.2, 3.20 & 4.4)

3.6 Marking. Marking shall conform to Specification MIL-M-13231 the equipment drawings. (See 3.2.2, 3.20 & 4.4)

3.6.1 Nameplate. Each Electronic Shop shall be sequentially serial numbered and provided with a nameplate in accordance with the drawings.

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3.6.2 Aircraft loading data plate. The shelter assemblage shall be provided with data plates in accordance with equipment drawings, imprinted with the dimensions which locate the center-of-gravity of the shelter facility and the assemblage in the three planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. The location of the centers-of-gravity shall be determined by the contractor using the first article/unit of production.

3.7 Riveting. Riveted joints shall be tight; the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. Rivets shall conform to requirement 12 of MIL-STD-454. (See 3.20)

3.8 Securing of parts. Brackets, lugs, flanges, inserts, bolts and other mounting arrangements shall retain the components of the Electronic Shops securely so that loosening, wear of mounting arrangements or permanent separation of parts or components will not occur when the equipment is subject to specified drop and transport tests. (See 3.20)

3.9 Soldering. (See 3.20)

3.9.1 Solder and flux. Solder and flux shall be as defined on the drawings,

3.9.2 Acid or acid salts. No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuts, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used as permitted above, they shall be completely neutralized and removed immediatley after use.

3.9.3 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment. Insulation material that has been subject to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

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3.10 Welding. Welded joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary except where grinding is specified on the drawings. Inert-gas-shielded arc welding using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 Cl. B, MIL-W-45205 Cl. B or MIL-W-8611 as applicable. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, adnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 3.2.2, 3.20 & 4.4)

3.11 Wiring and cabling. Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy. (See 3.2.2 & 3.20)

3.11.1 Slack. Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

c. To provide for at least two replacements of the part to which the wire or cable is connected, except for certain radio-frequency leads which must be as short as possible for electrical reasons.

d. To ensure freedom of motion of lugs or terminals normally intended to have some degree of movement.

e. To facilitate field repair of broken or cut wires.

f. To prevent chafing or breaking of individual wires due to repeated flexing of hinged parts.

3.11.2 Protection, Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.11.3 Splicing. Wires in a continous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

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3.11.4 Solderless-terminal-lug connections. Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements of MIL-T-7928. (See 4.4 & 4.11)

3.11.5 Identification of wiring. Wiring shall be identified by color coding and wire pair marking, as indicated on the drawings.

3.11.6 Cable assemblies. All telephone cord assemblies and air conditioner cable assembly shall be in accordance with Specification MIL-C-3885. (See 4.4)

3.12 Electrical installation. Power and signal wiring of each Electronic Shop shall be subjected to and pass the applicable tests described in 4.6. (See 3.20)

3.13 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.12) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.14 Tropicalization of material. Canvas, webbing, wood, etc. shall be treated to be resistant to moisture and fungi as indicated on the drawings and as follows: (See 3.20)

3.14.1 Treating materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material shall be held to the practical minimum.

3.14.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.14.3 Flexibility. Treatment shall not affect the flexibility of treated materils, to the extent that the equipment may fail to meet the specified requirements when subjected to specified service conditions.

3.14.4 Statement of treatment. The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materialsnd processes that he proposes to use. (See 6.2(f))

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3.15 Railroad transportation. The Electronics Shops shall be capable of being loaded, blocked and braced on a flatcar for shipment by rail. When so loaded, the equipment shall be subjected to and withstand the test of 4.7 without sustaining any permanent damage, such as buckling, crackling, delamination or other permanent deformation.

3.16 Drops, flat and rotational. The Electronics Shops shall withstand drops of 18 inches onto concrete. After each drop of 4.8.1 and 4.8.2 there shall be no buckling, cracking, delamination or other permanent damage to the shelter or its components.

3.17 Air Tightness Test. The shelter facility shall show no evidence of air leakage through seams, joints, rivets, etc. of modified exterior surfaces when tested for tightness in accordance with 4.9.

3.18 Interference Suppression. The motors in the blowers, air conditioners and heaters, the fluorescent lighting and the power supplies used in the Electronic Shops shall meet the emanation limits of MIL-STD-461 for tactical equipment, if other than those items defined by the drawings. (See 4.4 & 4.14)

3.19 Mounting inserts. All mounting inserts, unless otherwise specified, shall be installed without the use of any auxiliary materials such as adhesive and, when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.10. There shall be no damage or permanent distortion to the shelter mounting surface or insert head and threads and the inserts shall not have rotated from their original position. Insert threads shall be free of paint. (See 3.20)

3.20 Workmanship (See 4.16). The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.2	Construction
3.3	Seal, Water Vapor
3.4	Cleaning
3.5	Finish
3.6	Marking
3.7	Riveting
3.8	Securing of Parts
3.9	Soldering
3.10	Welding
3.11	Wiring & Cabling
3.12	Electrical Installation
3.14	Tropicalization
3.19	Mounting Inserts

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3.21 Systems Safety Engineering (See 4.13)

3.21.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. the criteria of MIL-STD-454, Requirement 1, shall be applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection. (See 4.13)

3.21.2 Edge Rounding Exposed. Exposed edges and corners shall be rounded sufficiently to minimize laceration/puncture hazards, the design goal being a minimum radius of 0.04 inch (1 mm) for edges and a minimum of 0.5 inch (12.7 mm) for corners.

3.21.3 Radioactive Materials. Radioactive materials shall not be used (e.g. luminous dials/markings, electron tubes, surge arrestor and lenses).

3.21.4 Exposed Voltage. All contacts, terminals and like devices having voltages between 70 and 500 volts rms or dc with respect to, ground, shall be guarded from accidental contact by personnel if such points are exposed to contact during direct support or operator maintenance.

3.21.5 Continuity of Conductors. The continuity of a grounded conductor shall not be dependent upon device connections, such as lampholders, receptacles, etc., where the removal of such devices would interrupt the continuity,

4.1.1 Government verification. All quality assurance operations required of the contractor shall be subject to Government verification at scheduled or unscheduled intervals. Verification will consist of the following:

(a) Surveillance of the contractor operations to determine that practices, methods, and procedures of the contractor’s product assurance system are being applied, and are in fact accomplishing the objectives of total compliance of the product with the requirements of this specification.

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4.1.2 Accommodation and Assistance. The Government quality assurance represenative shall have the right to access any area of the contractor's or his subcontractor's premises where any part of the work is being performed. The Government quality aussrance representative shall be afforded unrestricted opportunity to verify conformance of the product with specification requirements. The contractor shall make his inspection equipment available for use by Government quality assurance representatives for verification purposes. The contractors personnel shall be made available for operation of such inspection equipment as required. All facilities and services necessary for the placement, operation and maintenance of these test requirements shall be provided by the contractor.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (See 4.3) does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) packaging inspection. (See 4.17)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.1 & 6.3)

4.3.1 First article units. The contractor shall furnish one (1) first article unit of the complete AN/ASM-146( ) and one (1) first article unit of the complete AN/ASM-147( ).

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table I, and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

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TABLE I First Article Inspection

Inspection	Reqt Para	Test Para
Interchangeability	3,13	4.12
Soldering-terminal-lug connections	3.11.4	4.11
Mounting Insert	3.19	4.10
Inspection covered by subsidiary documents		4.4
Visual and Mechanical	3.3	4.13
Electrical installation	3.12	4.6
Air tightness	3.17	4.9
Railroad transportation	3.15	4.7
Flat drop	3.16	4.8.1
Rotational drop	3.16	4.8.2
Air tightness	3.17	4.9
Interference suppression	3.18	4.14
Visual and Mechanical	3.23	4.13
Electrical installation	3.12	4.6

4,4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this Specification , and the inspection requirement specified in the contract or purchase order.

Item	When Required
Finish	395
Marking	3.6
Welding	3.10
Cable Assemblies	3.11.6
Interference suppression	3.18
Solderless-terminal-lug connections	3.11.4
Riveting	3*7

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4 and 4.5.1 thru 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is deemed necessary to control the quality of the product and to assure compliance with all

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specification requirements. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records as an element of government verification. (See 4.1.1)

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Lots shall be formed only from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, AQL's indicated in Table II.

TABLE II Group A Inspection

Inspection	Reqt Para	Test Para	AQL% Major	Minor
Visual and Mechanical	3.23	4.13	1.5dphu*	6.5dphu**
Electrical installation	3.12	4.6	1.0	**
Air Tightness	3.17	4.9	4.0	**

*Defect per hundred unit
**All defects are major

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in Table II.

4.5.2 Group B inspection. Group B inspection shall be performed on lots that have passed Group A inspection. This inspection shall conform to Table III and the special inspection levels of Table I of MIL-STD-105.

TABLE III Group B Inspection

Inspection	Reqt Para	Test Para	AQL%
Interchangeability	3.13	4.12	4.0
Mounting inserts	3.19	4.10	4.0

4.5.2.1 Order of inspection within Group B. Group B Inspection shall be performed in the order listed in Table III.

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed Group A & B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1 thru 4.5.3.2.

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TABLE IV Group C Inspection

Inspection	Reqt Para	Test Para
Mounting inserts	3.19	4.10
Solderless-terminal-lug connections	3.11.4	4.11
Railroad transportation	3.15	4.7
Drops, flat and rotational	3.16	4.8

4.5.3.1 Sampling for inspection of solderless-terminal-lug connections. Five specimens of each combination of wire and terminal produced each month shall be prepared for inspection. A specimen is a one foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.2 Sampling for other Group C Inspection. For inspection tests listed below, one unit shall be randomly selected from the first 10 units (or fraction thereof) produced. Selection of the unit will be made by the Government. Since inspection of the mounting inserts needs to be performed prior to the completion of the unit, unit selected for mounting insert inspection shall, wherever possible, be different from the unit selected for other inspections.

a. Mounting inserts
b. Tropicalization of material
c. Interference suppression
d. Railroad transportation
e. Drops, flat and rotational

4.5.3.3 Order of inspection. Group C inspection shall be performed in the order listed in Table IV.

4.5.3.4 Non-compliance. All quality conformance inspection shall be halted, including group A & B inspections, upon occurrence of any group C failure. The contractor shall immediately report to the Government, in writing, each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on the processes and materials, as applicable on all units of production which are manufactured under the same conditions and which may be subject to the same failure. Reports shall be forwarded to the Government Procurement Quality representative and to the responsible technical activity designated in the contract.

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4.5.3.5 Reinspection of conforming group C Sample units. After corrective action has been taken, the failed unit and/or additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the technical activity). Any or all group A inspections may also be reinstituted. All damage resulting from any testing shall be repaired. Final acceptance and shipment will be withheld until the reinspection results have shown that the corrective action was effective and approved by the technical activity.

4.6 Electrical installation tests. (See 3.12)

4.6,1 Insulation breakdown test. (AN/ASM-146 & AN/ASM-147)

a. Remove all resistive and inductive loads from the lines such as power panel fuse, fluroescent, indancescent, and neon lamps, voltmeter and power plugs to equipments. Turn all breakers and switches to their ON positions except the lighting circuits and 120V A.C. bench receptacle Close the shelter door so that the blackout switch is activated.

b. Apply 100V DC Min to 500V DC Max test voltage for a period of ten (10) seconds between conductors and between each conductor and ground. The leakage resistance shall be no less than 50 megohms.

c. After test return all switches and breakers to their OFF positions and restore all resistive and inductive loads.

4.6.2 Test of lighting, circuit breaker #1 and door interlock switch. (AN/ASM-146 & AN/ASM-147).

a. Check power entrance box for proper power connection. Switch on main breaker. Voltmeter inside shelter should read input volts (approx. 120 VAC), and the neon light over the doorway should be on.

b. Switch on all the auxiliary breakers. Turn on the blackout by-pass switch and light #1 switch. Observe that half of the ceiling lights are illuminated.

c. Turn on light #2 switch. Observe that all the remaining ceiling lights are illuminated.

d. Turn off the blackout by-pass switch, with the shelter door opened, observe that all the lamps extinguish, and the neon light remain lit.

e. Close shelter door and turn latch to locked position. Observe that the door interlock switch causes all lamps to light again,

f. Unlatch and reopen door. The lamps extinguish once more as the latch is turned, but the neon lamp remains lit.

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g. Turn on blackout by-pass switch and leave on for remainder of inspection. All lamps are illuminated.

h. Turn off circuit breaker #1 with the lights on, and then turn the breaker on again. The lamps extinguish and then light up again.

4.6.3 Test of exhaust fans and circuit breakers #2 (AN/ASM-147 only).

a. Plug in both exhaust fans, and turn both fan switches on and off a few times. The fans start up and stop as their respective switches are turned on and off.

b, With the exhaust fans operating, turn off circuit breaker #2 and then back on again.

4.6.4 Test of electric heater #1 and breaker #3 & #8. (AN/ASM-147 & 146 respectively).

a. Connect the #1 electric heater's AC line plug into its proper receptacle. Observe that the heater blows air when its switch is set in "Fan" position, blows warm air after a few minutes in the "Heat" position (with thermostat turned up), and shuts off in the straight up position. Check that the thermostat controls heater operator by placing heater in "Heat" position, and turning thermostat setting down to colder temperature setting. A click will be heard and the shelters ammeter will show a drop of about 20 amps when the heater shuts off. The fan will continue to blow.

b. With the heater in one of its operating positions, turn off circuit breaker. Check that this shuts off the heater operation.

4.6.5 Test of convenience outlets and circuit breaker #4. (AN/ASM 147 only).

a. Plug test lamp into each convenience outlet, and observe that lamp illuminates.

b. With test lamp plugged into one of the outlets, throw off circuit breaker #4. Observe that the lamp extinguishes, and goes back on again when circuit breaker #4 is turned on again.

4.6.6 Test of electric heater #2 and circuit breaker #5. (AN/ASM-147). Connect the #2 electric heater's AC line plug into its proper receptacle and test as specified in 4.6.4. Repeat for circuit breaker #9 and heater #2 in AN/ASM-146,

4.6.7 Test of all circuit breakers and ammeter. (AN/ASM-146 & AN/ASM-147). Check that circuit breakers hold with all equipment operating in shelter (this includes electric heater). During this test, check to

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see that ammeter reads the approximate load applied.

4.6,8 Operational check of phone line. (AN/ASM-146 & AN/ASM-147). Check phone circuit. Connect two wires from one test phone to binding posts 1A and 1B in power entrance box. Connect tinned ends of telephone signal cord to the binding posts of the other test phone, and insert plug end into phone jack, roadside. Operate ringer on each phone, in turn, and talk into each phone. The phones should ring, and the conversation should be loud and clear.

4.6.9 Test of frequency converter, circuit breakers, and convenience outlets. (AN/ASM-146 only).

a. Connect the frequency converter's power line to its proper receptacle and turn the converter on.

b. With the converter operating, turn off the circuit breaker for the frequency converter and then back on again. The frequency converter shall stop and start again.

c. Using a frequency meter, check each convenience outlet for 400 cycles per second.

d. With the frequency meter connected to one of the outlets, throw the circuit breaker for the bench outlets to the off position. The frequency meter shall show no reading.

4.6.10 Test of Power Supply PP-1104( )/G, convenience outlets, and circuit breaker. (AN/ASM-146 only).

a. Connect the power supply's power line to its proper receptacle and turn the power supply on.

b. With the power supply operating, turn off circuit breaker for the power supply and then back on again. The power supply shall stop and start again.

co Using a D.C. voltmeter, check each convenience outlet for 28 volts D.C.

4.6.11 Test of air conditioner.

a. Open all air conditioning panel covers on the outside of the shelter.

b. Turn on breaker #2.

c. Turn air conditioner controller to "Ventilate". Air current shall flow from the ducts. Record current indicated on ammeter,

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d. Turn controller to "External" and thermostatic control to lowest temperature position. After a period of about five (5) minutes the temperature of the air from the air duct shall be cooler than the ambient air in the shelter. The unit shall function smoothly and ammeter reading shall have increased above the first observed reading.

e. Turn off breaker #2. Repeat for C.B. #4 and air conditioner #2.

4.6.12 Test of 400 cycle receptacles and circuit breakers. (AN/ASM-146 only).

a. Plug test lamp into each 400 cycle outlet and observe that lamp luminates.

b. With test lamp plugged into one of the 400 cycle outlets, throw off the 400 cycle circuit breaker. Observe that the lamp extinguishes, and goes back on again when the circuit breaker is turned on again.

4.7 Rail transport, impact test. The shelter assemblage with shelves loaded per 4.7.2, secured to a flatcar shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter tie down slings, shall be as specified in figure 1 and shall meet the provisions of the test as specified in 4.7.1. The test flatcar (less the shelter) and the two empty flatcars coupled to it shall each weigh between 45,000 and 55,000 pounds. The loaded gondola car shall meet the provisions specified in 4.7.3 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the Government Technical Representative) shall be impacted against the stationery loaded test car coupled to the empty flatcars with brakes off. The test shall consist of four impacts. The first two impacts shall be with the door end of the shelter facing the loaded gondola car; the second two impacts shall be with the door end of the shelter facing the opposite direction from the loaded gondola car. The lashing, blocking and wedges holding the shelter in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter assemblage shall meet the requirements of 3.15.

4.7.1 Test flatcar. The wooden planking of the test flatcar bed shall be in good condition. All 6x6 inch blocking shall be secured to the bed of the flatcar using 10-inch spikes on 2-inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles. This shall be accomplished by means of power sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings,

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along the full length of the castings. These four blocking members shall be notched or shimmed to contact the trim along the sides of the shelter. The 4x4 inch blocking shall be nailed to the bed of the flatcar as shown in figure 1. All other blocking shall be firmly secured as shown in figure 1. Lashing shall consist of four wire rope slings, each including a 3/8 inch diameter, 6x19, flexible steel, performed wire rope (having a breaking strength of 10,6OO pounds minimum), a turnbuckle, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9,000 pounds minimum. Before each impact, each sling shall be taut (600-800 pounds tension).

4.7.2 Shelter shelves shall be loaded with uniformly distributed dummy loads strapped to the shelves, centrally located, covering at least 50 percent of the shelf area. Shelf loading shall be as follows:

a. AN/ASM-146( ):
80 lbs each on each of the four shelves.

b. AN/ASM-147( ):

(1) Shelves of roadside racks starting from the bottom shelf: 180 lbs, 160 lbs, 100 lbs, 40 lbs, and 20 lbs.

(2) Shelves of curbside front rack starting from the shelf above the dehumidifier: 100 lbs, 70 lbs, and 20 lbs.

(3) Shelves of curbside rear racks starting from the bottom shelf: 180 lbs, 60 lbs, 50 lbs, and 20 lbs.

4.7.3 Loaded gondola car. The load in the loaded gondola car shall be of the following:

a. A coarse size of either coal, crushed stone, gravel, concrete or metal scrap (other than sheet metal, metal shavings or other springy forms of metal).

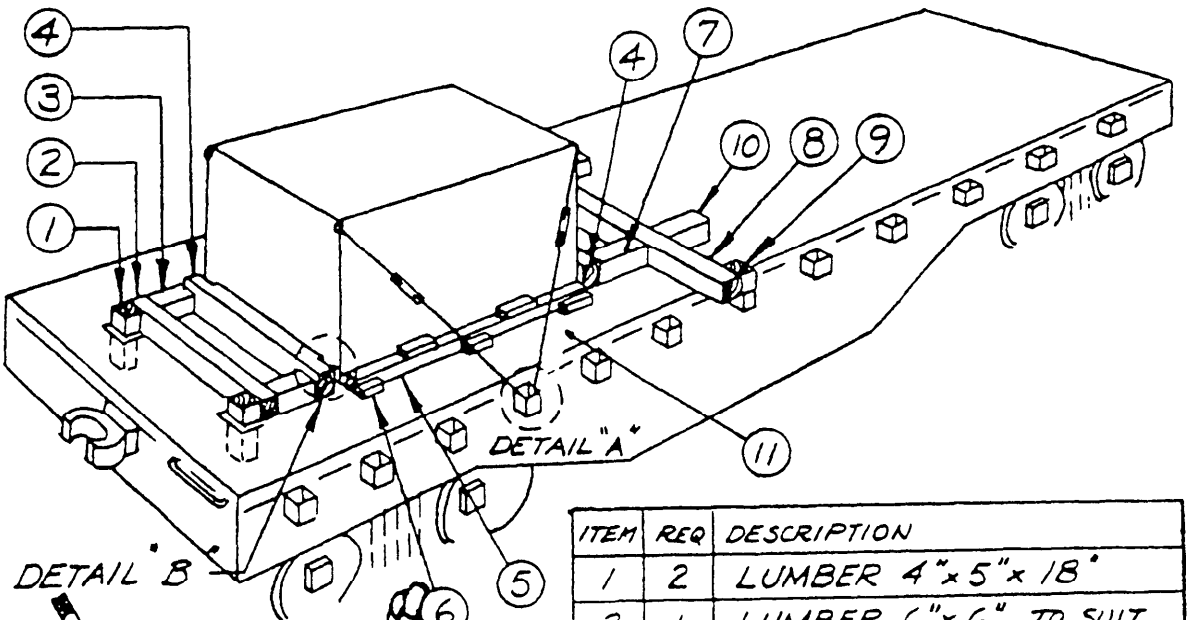
b. Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting,

4.8 Drop test

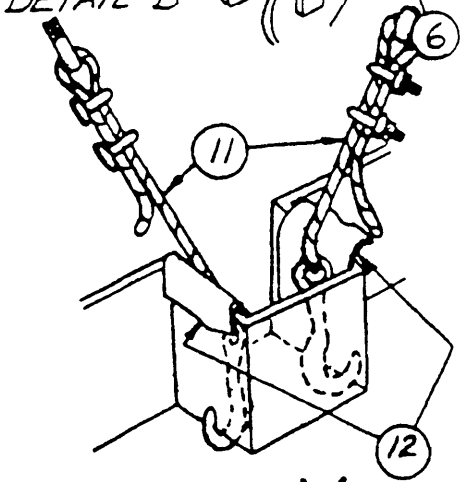
4.8.1 Flat drop. The shelter assemblage shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the assemblage shall meet the requirements of 3.5.

4.8.2 Rotational Drop. The shelter assemblage shall have one skid (or the knuckles of the skids) resting on a nominal 4-inch high timber. The outer edge of the opposite skid (or the opposite knuckles of the

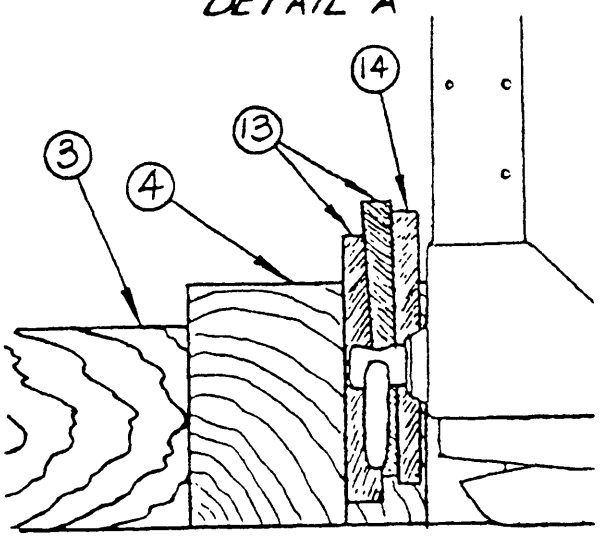
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DETAIL "B"



DETAIL "A"



DETAIL "B"

ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"x 5"x 18"
2	1	LUMBER 6"x 6" TO SUIT
3	2	LUMBER 6"x 6" TO SUIT
4	2	LUMBER 8"x 8" TO SUIT
5	2	LUMBER 4"x 4" x 147"
6	10	LUMBER 2"x 4" x 14"
7	2	LUMBER 6"x 6" TO SUIT
8	1	LUMBER 6"x 6" TO SUIT
9	2	LUMBER TO SUIT
10	2	LUMBER 6"x 6"x 24"
11	4	TIEDOWN SLINGS OF 3/8" 6X19 WIRE ROPE (10,600 POUNDS MINIMUM BREAKING STRENGTH). SEE 4.6.21.1
12	4	FLEX. METAL SHEETING
13	8	OAK WEDGES, 8"x 3"x 1" and TAPER TO 1/4" thk TWO WEDGES ARE REQ'D TO ASSURE PROPER BEARING BTWN CORNER CASTING & BLOCKING
14	4	OAK SHIMS, 8"x 3"x 1" SEE DETAIL "B"

FIGURE 4 -
SHELTER SECURED ON FLATCAR

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skids) shall be raised 18 inches and the shelter shall be allowed to fall freely onto a hard substantial concrete surface. This inpact shall be performed four times, once for each bottom edge of the shelter. On completion of these drops, the shelter assemblage shall meet the requirements of 3.16.

4.9 Air tightness test. The shelter facility when subjected to the following test, shall meet the requirements of 3.16.

a. The air tightness shall be tested by introducing clean dry air into the closed shelter through an existing appropriate opening (i.e. drain fixture) to create an internal air pressure that exceeds the external pressure by a minimum of six (6) inches of water.

b. A manometer graduated in 0.1 inches or less shall be utilized to measure this differential pressure.

c. The test apparatus shall include an air supply sufficient to maintain the required air pressure, an air flow regulator and an input air pressure gage.

d. The shelter facility shall be tested for air tightness by painting the sealed joints, seams, rivets, etc. of the external modifications with Leak-Tat Formula 372 (American gas & Chemicals, Inc., Leak-Tat Division) or equal soap and water solution.

e. To minimize air loss, temporary seals, shims and tape are permitted to seal the shelter door, blower cover and air conditioner port openings not being tested.

4.10 Mounting insert tests. Four of each size inserts shall be randomly selected for test. Each insert shall be tested with an axial load followed by a torque load in accordance with Table V. After test, the requirements of 3.19 shall be met. Failure of any insert shall be cause for rejection of the sample equipment. This test shall be conducted prior to installation of brackets, racks, etc.

TABLE V

Mounting insert loads

Insert Size	First Article and Group C Test Axial (lbs)	Torque (Inch Lbs)	Group B Test Axial (Lbs)	Torque (Inch lbs)
I 0-32	600	23	300	23
1/4-20	1300	60	650	60
5/16-18	2000	100	1000	100

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4.11 Testing of solderless terminal-lug connections. Connections made with solderless terminal-lugs shall be tested for conformance to 3.11.4.

4.12 Inspection for interchangeability. The dimensions listed below shall be gauged or measured to determine compliance with the physical interchangeability requirements of 3.13. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks and cabinets.

4.13 Visual and mechanical inspection. The shelter assemblage shall be examined for the defects listed in MIL-STD-252, the requirements of 3.23 and TABLE VI. An inspection shall also be performed to verify compliance with those portions of 3.21 (Systems Saftey Engineering) which can be determined visually.

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TABLE VI

Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or misloated. 2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible. 3* Welds-cracked or porous. Inspection of welded assemblies, such as equipment racks entrance boxes, etc. should be made prior to final paint application. 4. Sealer improperly applied (holes, separations, or lack of adhesion). 5. Mounting inserts missing or dislocated. 6. Burrs or sharp edges in wire duct not removed. 7. Equipments or bracketry loose or missing. 8. Doors, etc. inoperative. 99 Electrical receptacles, switches, jacks, connectors, wire ducts, or other electrical components loose, improperly located or inoperative. 10. Locking or holding devices missing or inoperative. 11, Gaps between sections of wire duct exceed 1/16 inch. 12. The blower housing or air conditioner mount does not align properly with the mounting inserts of the shelter facility. 13. Ground stud not properly sealed. 14. Bulkhead power and signal connector. 15. Electrical schematics missing.
Minor	1. Doors not easily operable. 2. Fastening devices difficult to operate. 3. Finish-abrasions or scratches. 4. Burrs and sharp edges on brackets not removed. 5. Screws loose. 6. Gaskets, not completely and securely adhered. 7. Electrical schematics not properly adhered to wall. 8. Cable assemblies not dressed properly.

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4.14 Electromagnetic interference test. The items stated in 3.18 shall be determined to comply with the requirements of 3.18. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, as implemented by a contractor prepared, Government approved, EMI Test Plan.

4.15 Cable Assemblies, power and audio frequency. Each cable assembly shall be tested for continuity, dielectric, insulation resistance and pull-out per MIL-C-3885. (See 3.11.6)

4.16 Tropicalization of materials. All materials of construction shall be in accordance with the equipment drawings and shall be Fungus-inert. Substitution of any materials other than specified must be approved by the contracting officer. The material will only be approved after the contractor tests the material for Fungus-Resistance satisfactorily in accordance with the requirements 508.1 of MIL-STD-81Oc.

4.17 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.

b. Quality Conformance Inspection of Packaging.

4.17.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.17.2 Quality Conformance Inspection of Packaging.

4. 17.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4. 17.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirement of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4. 17.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in the Packing Inspection Provisions Table. Lot formation shall consist of all packs made of the same materials during an identifiable period and-submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective.

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TABLE VII Packing Inspection Provisions

No.	Characteristic	Method of Inspection
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measurement
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

5. Packaging.

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-55507.

6. Notes

6.1 Intended use. Shops, Electronic, Shelter Mounted, AN/ASM-146( ) and AN/ASM-147( ) are used at Army aviation division or battalion centers to provide backup maintenance support for the electronic equipment components installed in aircraft. Specific use is as follows:

a. Shop, Electronic, Shelter Mounted, AN/ASM-146( )is used as the repair or maintenance shop for accomplishing the maintenance functions.

b. Shop, Electronic, Shelter Mounted, AN/ASM-147( ) is used as the storage facility for the repair parts and electronic equipment major components maintenance float.

6.2 Ordering data. Procurement documents should specify the following:

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a. Title, number, and date of this specification and any amendment thereto.

b. Shop type required.

c. Level A or B preservation and packing (see 5.1).

d. Marking and shipping of samples.

e. Place of final inspection.

f. Submission of the statement of treatment referenced in 3.14.4 as soon as possible after award of contract and before proceeding with the work. This statement shall be submitted to the contracting officer.

g. When first article rough handling tests are not required (see 4.17).

h. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packaging.

6.2 First Article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample. The first article should consist of one each AN/ASM-146( ) and one each AN/ASM-147( ) units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article. (See 4.3 and 3.1)

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.4 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.2.1.

Custodian
Army-CR

Preparing Activity
Army-CR

Project No. 4940-A493

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-55257C (CR)	SHOPS, ELECTRONIC, SHELTER MOUNTED AN/ASM-146( ) & AN/ASM-147( )

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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